Practical 22: Runtime Permissions and Security

Source: MODULE 2.pdf | Source pages: 89–94

Aim

To develop and test an Android application demonstrating runtime permissions and security.

Steps to Perform the Practical

1. Create an Android Studio project for runtime permissions.

2. Review the permission and security requirement in the source practical.

3. Declare the required permission in AndroidManifest.xml when applicable.

4. Check permission status at runtime and request permission when it has not been granted.

5. Handle both granted and denied results without crashing.

6. Run the app, test the permission prompt, and verify the app's behavior for both Allow and Deny.

Result

The application is run and the expected behavior is verified according to the practical instructions and code.

Note: Use the package name and resource names from the source code consistently. If Android Studio reports a missing resource or class, check spelling and ensure all files are in the correct folders.

15. Develop an Android application implementing Runtime Permissions and security handling in AndroidManifest.xml

Project Name:PermissionDemo

Add Permission in AndroidManifest.xml.
Open:
AndroidManifest.xml
Add below code lines above: <application> tag
<uses-permission android:name="android.permission.CAMERA"/>
<uses-feature android:name="android.hardware.camera"
android:required="false" />

```
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools">
    <uses-permission android:name="android.permission.CAMERA"/>
    <uses-feature android:name="android.hardware.camera"
        android:required="false" />

    <application
        android:allowBackup="true"
        android:dataExtractionRules="@xml/data_extraction_rules"
        android:fullBackupContent="@xml/backup_rules"
        android:icon="@mipmap/ic_launcher"
```

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center">
    <Button
        android:id="@+id/btnPermission"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
```

```
        android:text="Request Camera Permission"/>

</LinearLayout>
```

MainActivity.java

```
package com.example.permissiondemo;


import android.content.pm.PackageManager;

import android.os.Bundle;

import android.widget.Button;

import android.widget.Toast;

import androidx.appcompat.app.AppCompatActivity;

import androidx.core.app.ActivityCompat;

import androidx.core.content.ContextCompat;


public class MainActivity extends AppCompatActivity {


    Button btnPermission;


    @Override

    protected void onCreate(Bundle savedInstanceState) {

        super.onCreate(savedInstanceState);

        setContentView(R.layout.activity_main);


        btnPermission = findViewById(R.id.btnPermission);
```

```
    btnPermission.setOnClickListener(v -> {
        checkPermission();
    });
}

private void checkPermission() {

    // Check if Camera permission is already granted
    if (ContextCompat.checkSelfPermission(
            this,
            android.Manifest.permission.CAMERA
    ) == PackageManager.PERMISSION_GRANTED) {

        Toast.makeText(
                this,
                "Permission Already Granted",
                Toast.LENGTH_SHORT
        ).show();

    } else {

        // Request Camera permission
        ActivityCompat.requestPermissions(
                this,
                new String[]{
                        android.Manifest.permission.CAMERA
```

```
                },
                1
        );
    }
}

// Permission Result
@Override
public void onRequestPermissionsResult(
        int requestCode,
        String[] permissions,
        int[] grantResults
) {
    super.onRequestPermissionsResult(
            requestCode,
            permissions,
            grantResults
    );

    if (requestCode == 1) {

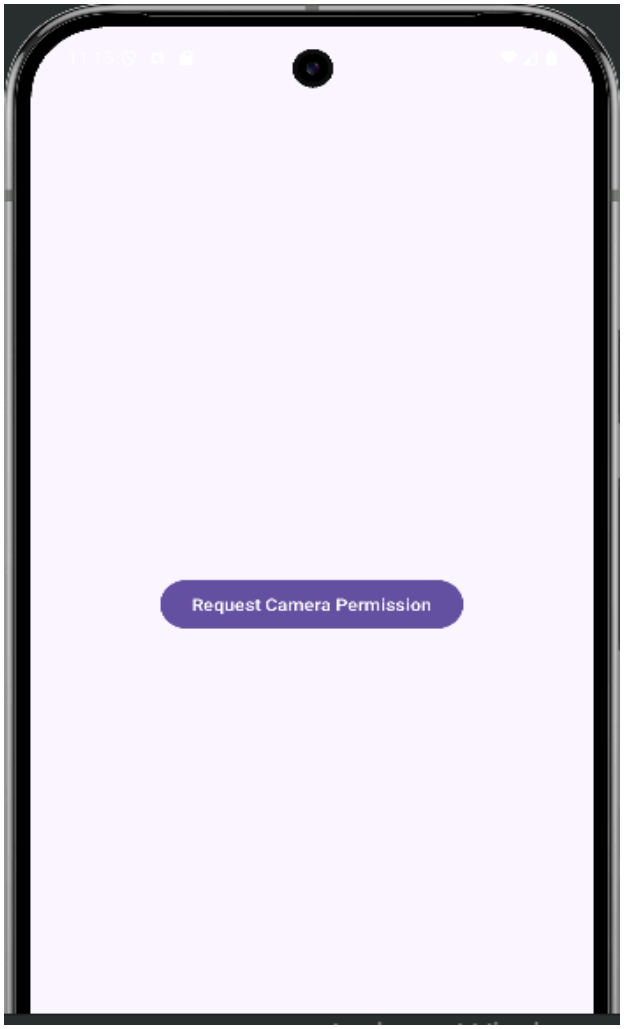
        if (grantResults.length > 0
                && grantResults[0] == PackageManager.PERMISSION_GRANTED) {

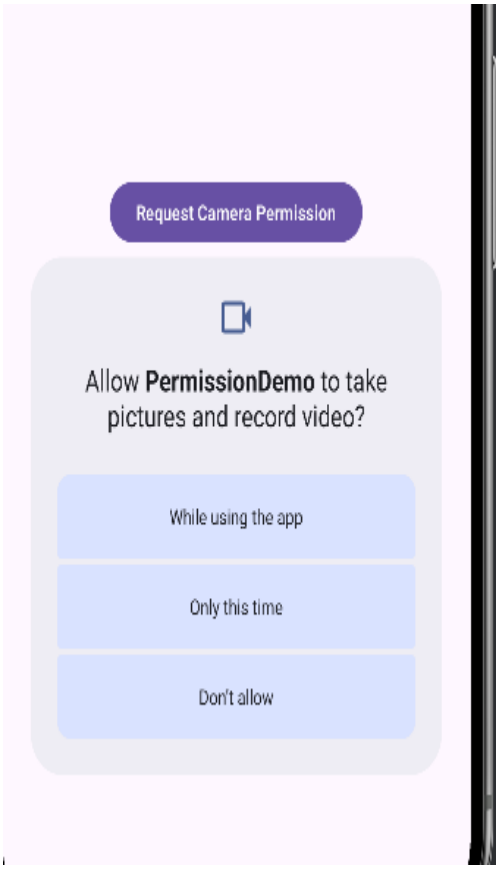
            Toast.makeText(
                    this,
```

```
                        "Permission Granted",
                        Toast.LENGTH_SHORT
                ).show();

            } else {

                Toast.makeText(
                        this,
                        "Permission Denied",
                        Toast.LENGTH_SHORT
                ).show();
            }
        }
    }
}
```


Request Camera Permission


Request Camera Permission
Allow PermissionDemo to take pictures and record video?
While using the app
Only this time
Don't allow